

Quantenpraktikum - NV Centre ODMR Experiments

22.01.2026

Authors: Daniel Köstler
Nico Machat
Michael Schleinzer

Contents

1	Experimental Setup	2
2	Theoretical Background	4
2.1	ODMR and Zero-Field Splitting	4
2.2	Zeeman Effect and Magnetic Field Sensitivity	4
2.3	Hyperfine Interaction	4
2.4	Determination of Weak and Strong Magnetic Fields	5
3	Experiments	6
3.1	Optically Detected Magnetic Resonance (ODMR)	6
3.1.1	Results of the ODMR measurements	7
3.2	Observation of Zeeman Splitting	8
3.2.1	Observed Zeeman Splitting	8
3.3	Gyromagnetic Ratio	10
3.3.1	Frequency Splitting and Gyromagnetic Ratio	10
3.3.2	Uncertainty Analysis	12
3.4	Nonlinear Contribution to the Zeeman Splitting	13
3.4.1	Nonlinear Frequency Shift	13
3.4.2	Uncertainty Analysis	15
3.5	Hyperfine Structure of the NV Centre	16
3.5.1	Observed Hyperfine Structure	16
3.5.2	Uncertainty of the Hyperfine Analysis	19
3.6	Determination of the Earth's Magnetic Field	20
3.6.1	Measured Earth's Magnetic Field	20
3.6.2	Uncertainty of the Earth's Magnetic Field Measurement	21
3.7	Strong Magnetic Field	22
3.7.1	Observed ODMR response in a strong magnetic field	22
4	Appendix	24
	List of Figures	24
	List of Tables	24

1 Experimental Setup

The experiments were performed using the *KWANTEACH*[®] NV center education kit by KWAN-TEK. The setup combines an optical confocal microscope, microwave excitation, and photoluminescence detection to study the spin properties of nitrogen-vacancy (NV) centers in diamond under ambient conditions.

A continuous-wave green laser (wavelength ≈ 532 nm) is used to optically excite the NV centers. The laser beam is directed towards the diamond sample using a set of mirrors and focusing optics and is tightly focused onto the sample by a microscope objective. The diamond is mounted at the center of a radio-frequency (RF) antenna, which is used to apply microwave radiation for spin manipulation.

The photoluminescence emitted by the NV centers in the red spectral range is collected by the same objective and separated from the excitation light using appropriate optical filters. The filtered fluorescence signal is then detected by a photodetector and recorded by the control software.

Microwave signals with frequencies around 2.87 GHz are generated and delivered to the RF antenna. By sweeping the microwave frequency while continuously monitoring the photoluminescence intensity, optically detected magnetic resonance (ODMR) spectra are obtained. External magnetic fields can be applied along the NV axis to lift the degeneracy of the spin states and induce a Zeeman splitting of the resonance lines.

All measurements are performed at room temperature and are controlled via dedicated software provided with the experimental system.

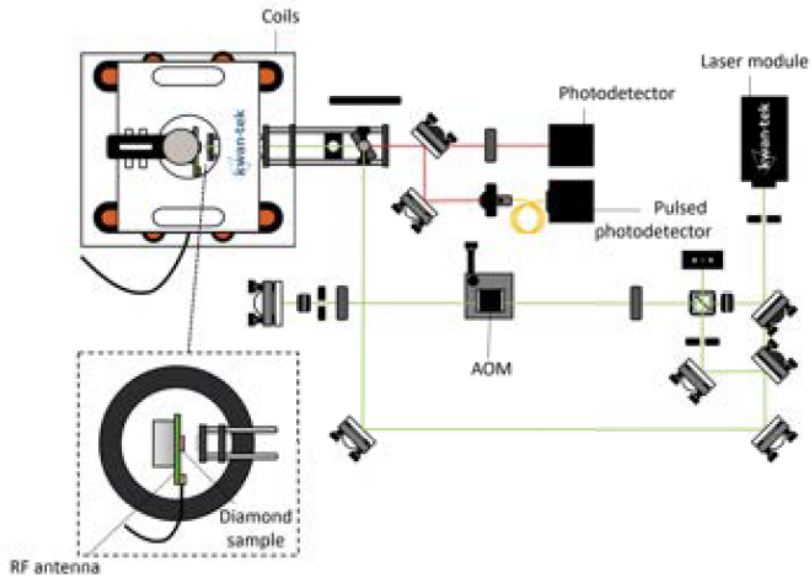


Figure 1: Optical bench configuration used for the experiments on NV centers

2 Theoretical Background

The nitrogen-vacancy (NV) centre in diamond consists of a substitutional nitrogen atom adjacent to a lattice vacancy. In its negatively charged state, the NV centre exhibits an electronic spin triplet ground state ($S = 1$), which can be initialized and read out optically. Optical excitation leads to spin-dependent photoluminescence, allowing the spin state to be detected via optically detected magnetic resonance (ODMR).

2.1 ODMR and Zero-Field Splitting

In the absence of an external magnetic field, the ground-state spin levels of the NV centre are split by the crystal field interaction. This so-called zero-field splitting separates the $m_s = 0$ state from the degenerate $m_s = \pm 1$ states by an energy corresponding to a frequency of approximately $D = 2.87$ GHz. When microwave radiation resonant with this splitting is applied, transitions between the spin states occur, leading to a reduction of the photoluminescence signal. These transitions appear as characteristic dips in the ODMR spectrum.

2.2 Zeeman Effect and Magnetic Field Sensitivity

An external magnetic field lifts the degeneracy of the $m_s = \pm 1$ states via the Zeeman interaction. For a magnetic field component projected onto the NV axis, the resonance frequencies are shifted according to

$$f_{\pm} = D \pm \gamma B_{\parallel}, \quad (1)$$

where γ denotes the gyromagnetic ratio of the NV centre and $B_{\parallel}$ is the magnetic field component along the NV axis. The resulting frequency splitting

$$\Delta f = f_{+} - f_{-} = 2\gamma B_{\parallel} \quad (2)$$

forms the basis for magnetic field sensing with NV centres.

For sufficiently small magnetic fields, the frequency shifts depend linearly on the applied field. At higher field strengths, additional contributions lead to deviations from linear behaviour, which can be described by higher-order terms in the magnetic field.

2.3 Hyperfine Interaction

In addition to the Zeeman interaction, the electronic spin of the NV centre couples to the nuclear spin of the host nitrogen atom. This hyperfine interaction leads to a further splitting of each ODMR transition into three closely spaced resonance lines, corresponding to the nuclear spin projections $m_I = -1, 0, +1$. The hyperfine splitting is on the order of a few megahertz and is independent of the applied magnetic field.

The linewidth and contrast of the hyperfine resonances depend on experimental parameters such as the applied microwave power. Increasing the microwave power leads to power broadening of the resonance lines and affects the observed ODMR contrast.

2.4 Determination of Weak and Strong Magnetic Fields

By measuring the frequency shifts of the ODMR resonances, both weak and strong magnetic fields can be detected. For weak magnetic fields, the projection of the field onto the NV axis can be determined from the frequency splitting of the two ODMR resonances according to

$$B_{\parallel} = \frac{\Delta f}{2\gamma}, \quad (3)$$

where $\Delta f = f_+ - f_-$ denotes the measured frequency splitting and γ is the gyromagnetic ratio of the NV centre.

Weak static fields, such as the Earth's magnetic field, can be extracted by comparing ODMR spectra recorded for opposite bias field directions. Reversing the bias field changes the sign of the coil-generated contribution, while the Earth's magnetic field remains unchanged, which allows its projection onto the NV axis to be isolated.

In the presence of strong magnetic fields, the ODMR spectrum is significantly distorted and individual resonances may no longer be clearly resolved. This qualitative behaviour illustrates the wide dynamic range of NV-centre-based magnetic field sensing.

3 Experiments

3.1 Optically Detected Magnetic Resonance (ODMR)

Optically detected magnetic resonance (ODMR) measurements were performed to probe the spin transitions of the NV centres in diamond. The experiment relies on continuous optical excitation combined with microwave-driven spin manipulation while monitoring the photoluminescence signal.

During the measurement, the NV centres were continuously excited using a green laser operated at 35% of its maximum power. At the same time, a microwave signal was applied via the RF antenna. The microwave frequency was swept in the range from 2.80 GHz to 2.94 GHz while the photoluminescence intensity was recorded.

When the microwave frequency matches the transition between the $m_s = 0$ and $m_s = \pm 1$ spin states, a reduction in photoluminescence is observed. This results in characteristic resonance dips in the recorded spectrum. The ODMR spectrum therefore provides direct access to the spin resonance frequencies of the NV centres.

3.1.1 Results of the ODMR measurements

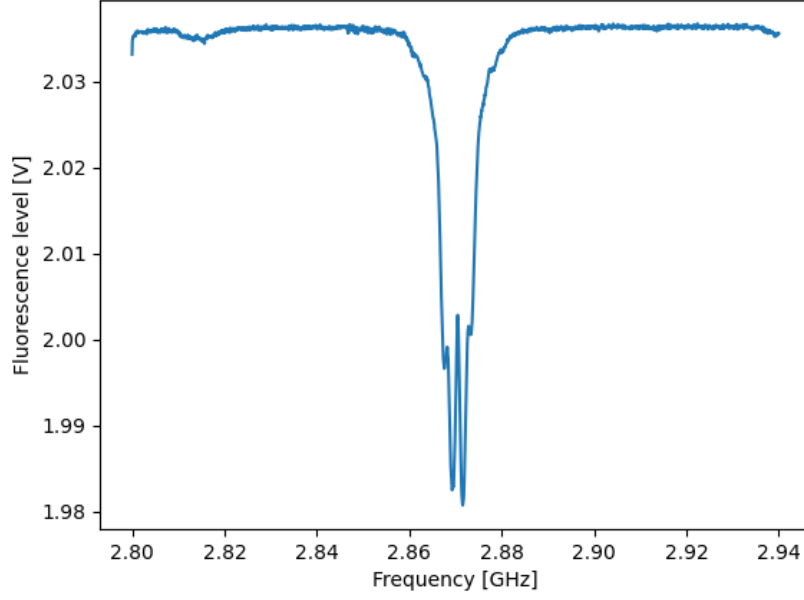


Figure 2: ODMR spectrum recorded at 35% laser power.

A clear resonance dip centered around 2.87 GHz was observed. This dip corresponds to the transition between the $m_s = 0$ and $m_s = \pm 1$ spin states of the NV centres. The measurement confirms the presence of optically detectable magnetic resonance and serves as a reference for the subsequent experiments.

3.2 Observation of Zeeman Splitting

ODMR spectra were recorded in the presence of an external magnetic field generated by pairs of coils. Initially, the magnetic field was not aligned with a single NV axis. In this configuration, an ODMR spectrum exhibits eight distinct resonance dips, corresponding to the four crystallographic NV orientations in the diamond.

The magnetic field was then aligned along one NV axis. After alignment, an ODMR spectrum shows four resonance dips, originating from the remaining NV orientations with a finite projection of the magnetic field. This aligned configuration was used for all subsequent Zeeman splitting measurements.

3.2.1 Observed Zeeman Splitting

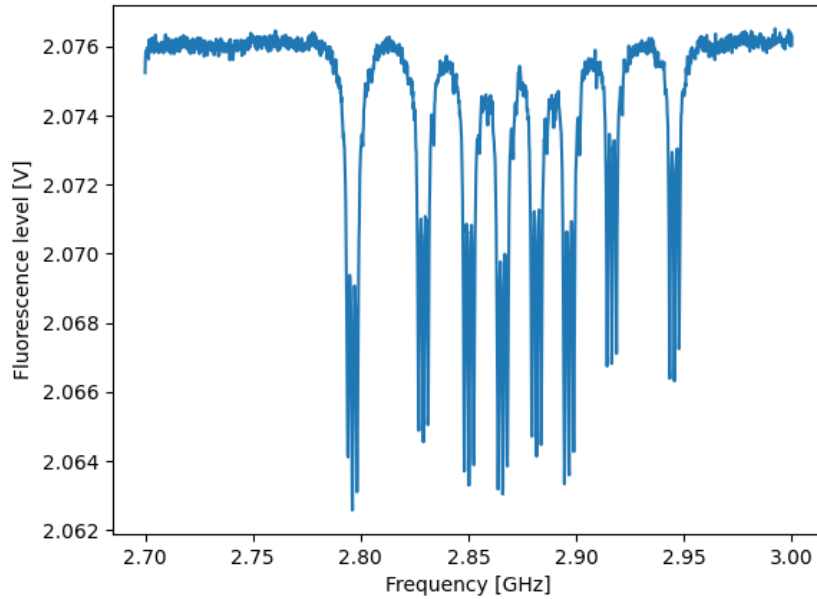


Figure 3: ODMR spectrum recorded in the presence of an external magnetic field not aligned with a single NV axis

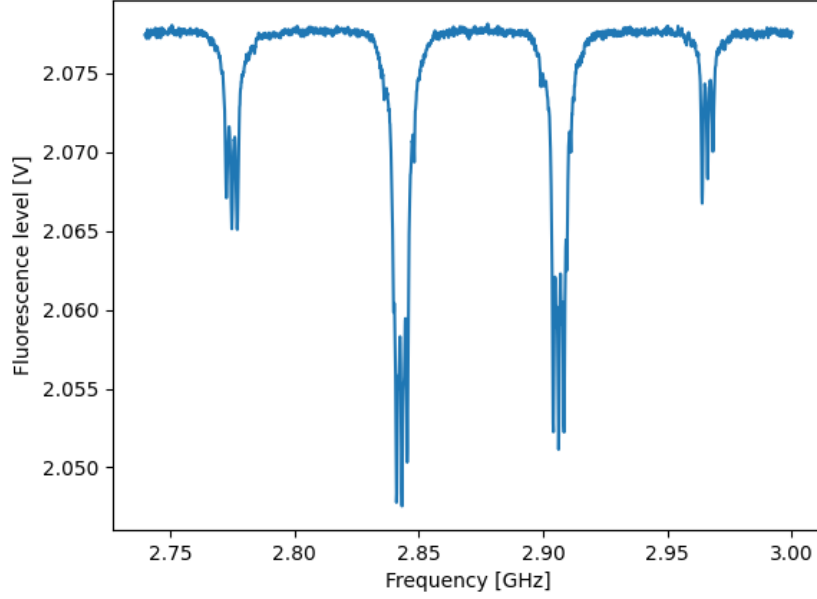


Figure 4: ODMR spectrum recorded with the magnetic field aligned along one NV axis

As expected, the ODMR spectra recorded with a non-aligned magnetic field exhibit eight resonance dips, corresponding to the four crystallographic NV orientations in the diamond. After alignment of the magnetic field along a single NV axis, only four resonance dips remain visible in the spectrum. This observation is consistent with the projection of the magnetic field onto the different NV orientations.

3.3 Gyromagnetic Ratio

For the determination of the gyromagnetic ratio, ODMR spectra were recorded under different applied magnetic field strengths. The magnetic field was generated using pairs of coils aligned along a single NV axis, as described in the previous section.

Measurements were performed for several fixed magnetic field values in the range from 1.8 mT to 3.4 mT. For each magnetic field setting, the microwave frequency was swept around the NV centre resonance while the photoluminescence signal was recorded.

All spectra were acquired under identical experimental conditions, with constant laser power and microwave power, in order to ensure comparability between the measurements.

3.3.1 Frequency Splitting and Gyromagnetic Ratio

The resonance frequencies f_+ and f_- were extracted from ODMR spectra recorded at different magnetic field strengths. Figure 6 shows the dependence of the two resonance frequencies on the applied magnetic field.

Both f_+ and f_- exhibit a linear dependence on B , with opposite slopes as expected for Zeeman-split spin transitions. Linear fits were applied to both data sets to determine the corresponding slopes.

From the fits, the gyromagnetic ratios γ_+ and γ_- were obtained. The mean value

$$\gamma = \frac{\gamma_+ + \gamma_-}{2}$$

was used as the experimental estimate of the gyromagnetic ratio of the NV centre.

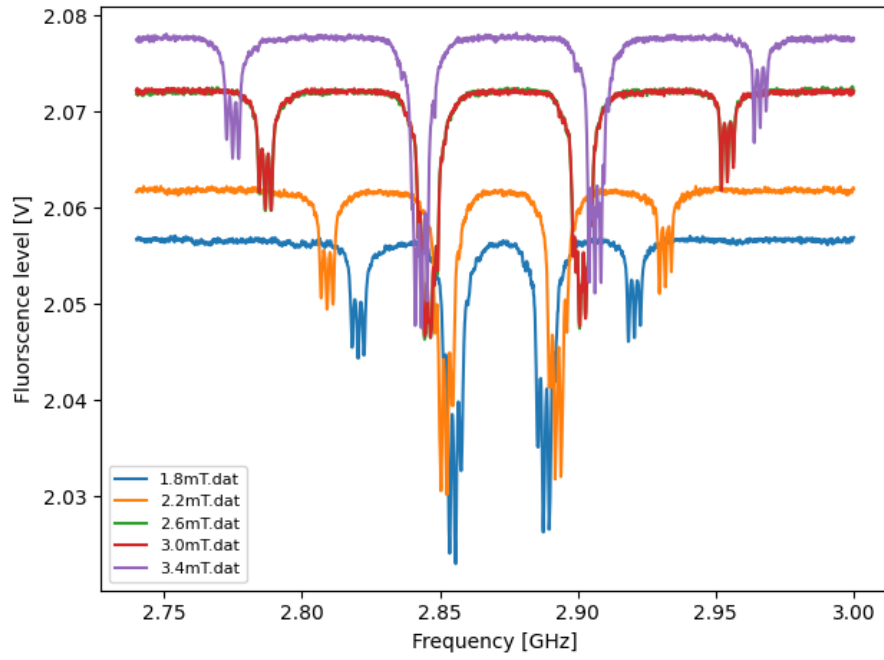


Figure 5: ODMR spectra recorded for different magnetic field strengths B (all measurement files)

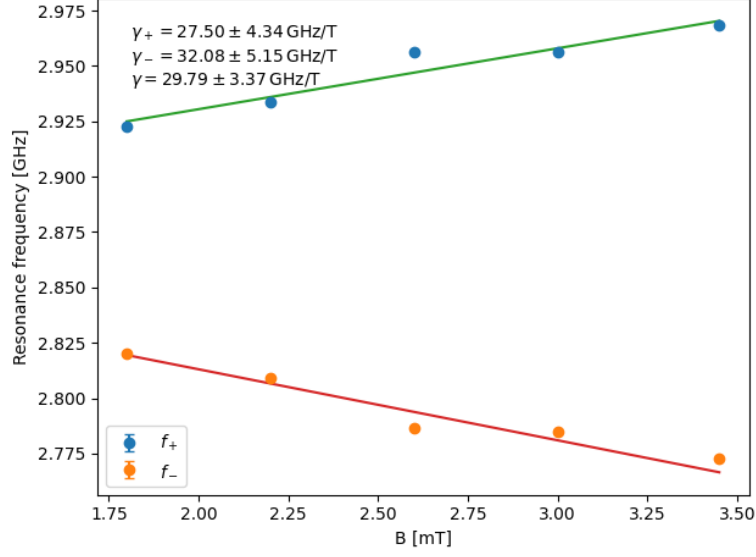


Figure 6: Resonance frequencies f_+ and f_- as a function of the applied magnetic field B . Solid lines indicate linear fits used to extract the gyromagnetic ratio

B [mT]	f_- [GHz]	f_+ [GHz]	Δf [MHz]	Resonance dips
1.80 ± 0.004	2.82033 ± 0.00007	2.92265 ± 0.00010	102.31 ± 0.24	7
2.20 ± 0.004	2.80912 ± 0.00007	2.93389 ± 0.00009	124.77 ± 0.27	7
2.60 ± 0.005	2.78672 ± 0.00008	2.95635 ± 0.00010	169.63 ± 0.36	6
3.00 ± 0.006	2.78461 ± 0.00013	2.95639 ± 0.00009	171.78 ± 0.37	8
3.45 ± 0.007	2.77280 ± 0.00014	2.96827 ± 0.00009	195.47 ± 0.42	8

Table 1: Extracted resonance frequencies and frequency splittings for different applied magnetic field strengths

The calculated value for the gyromagnetic ratio turns out to be $29.79 \pm 3.37 \text{ GHz/T}$ which matches well with the theoretical value of $29.4 \pm 1.2 \text{ GHz/T}$.

3.3.2 Uncertainty Analysis

The uncertainties of the resonance frequencies f_- and f_+ originate from the Lorentzian fits used to determine the dip positions and represent statistical uncertainties derived from the fit covariance matrices. The uncertainty of the frequency splitting Δf is dominated by the combined fit uncertainties of the two independently determined resonance frequencies.

In addition to these statistical contributions, systematic uncertainties arise from the determination of the applied magnetic field. The magnetic field is generated

by three orthogonal pairs of coils and is inferred from the applied coil currents and calibration factors rather than from a direct magnetic field measurement. According to the experimental manual, the relative uncertainties of the magnetic field generated by the individual coil axes are

$$\frac{\Delta B_x}{B_x} \approx 0.17\%, \quad \frac{\Delta B_y}{B_y} \approx 0.18\%, \quad \frac{\Delta B_z}{B_z} \approx 0.26\%.$$

Since the experiments are evaluated in terms of the magnitude of the effective magnetic field rather than its individual Cartesian components, a single representative uncertainty was adopted. The relative uncertainty of the applied magnetic field was therefore estimated using the arithmetic mean of the individual coil uncertainties,

$$\frac{\Delta B}{B} = \frac{1}{3} \left(\frac{\Delta B_x}{B_x} + \frac{\Delta B_y}{B_y} + \frac{\Delta B_z}{B_z} \right) \approx 0.20\%.$$

This averaged relative uncertainty was used consistently for all field-dependent quantities and propagated accordingly.

3.4 Nonlinear Contribution to the Zeeman Splitting

To investigate deviations from the linear Zeeman behaviour, ODMR spectra were recorded at several fixed magnetic field strengths in the range from 1.2 mT to 2.4 mT. The magnetic field was not aligned along a single NV axis.

For each magnetic field setting, the microwave frequency was swept in a narrow interval around the zero-field splitting frequency of approximately 2.87 GHz. The two resonance dips closest to the central frequency were selected for further analysis.

The frequency shifts of these resonances were evaluated as a function of the applied magnetic field in order to identify potential nonlinear contributions to the Zeeman splitting.

3.4.1 Nonlinear Frequency Shift

The resonance frequencies closest to the zero-field splitting were evaluated for several magnetic field strengths between 1.2 mT and 2.4 mT. Figure 7 shows the corresponding ODMR spectra in the vicinity of 2.87 GHz.

The extracted resonance frequencies f_+ and f_- exhibit a clear deviation from a purely linear dependence on the magnetic field. To account for this behaviour, the data were fitted using a second-order polynomial in the magnetic field.

The resulting fits reveal a measurable quadratic contribution to the magnetic-field dependence of the resonance frequencies, indicating the presence of a non-linear Zeeman term in this field regime.

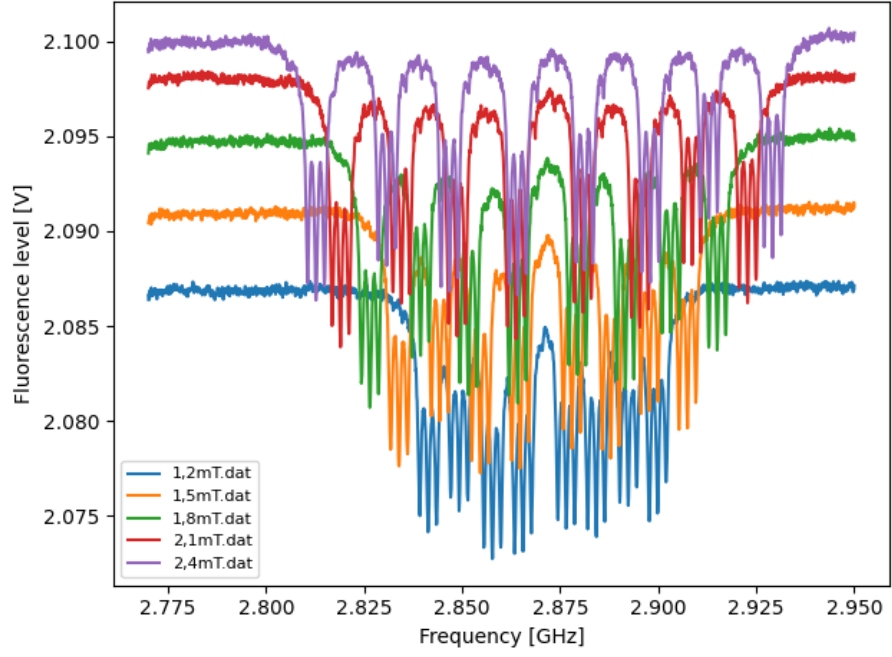


Figure 7: ODMR spectra recorded at different magnetic field strengths in a narrow frequency range around the zero-field splitting

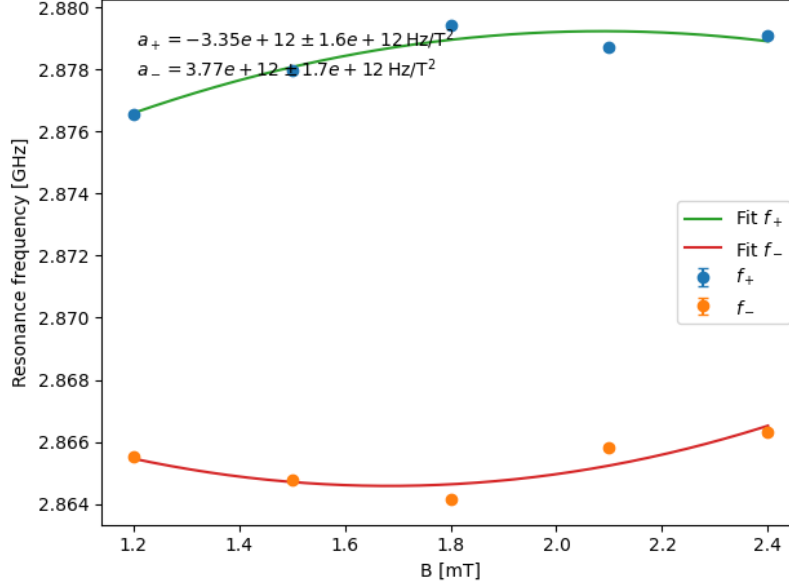


Figure 8: Resonance frequencies f_+ and f_- as a function of the applied magnetic field B

B [mT]	f_- [GHz]	f_+ [GHz]	D [GHz]	E [MHz]
1.20 ± 0.002	2.86552 ± 0.00006	2.87656 ± 0.00005	2.87104 ± 0.00004	5.517 ± 0.046
1.50 ± 0.003	2.86476 ± 0.00010	2.87798 ± 0.00006	2.87137 ± 0.00004	6.607 ± 0.050
1.80 ± 0.004	2.86416 ± 0.00005	2.87942 ± 0.00005	2.87179 ± 0.00004	7.628 ± 0.054
2.10 ± 0.004	2.86583 ± 0.00008	2.87869 ± 0.00007	2.87226 ± 0.00006	6.432 ± 0.064
2.40 ± 0.005	2.86631 ± 0.00008	2.87909 ± 0.00007	2.87270 ± 0.00005	6.390 ± 0.066

Table 2: Extracted resonance frequencies near the zero-field splitting and derived parameters for the nonlinear Zeeman analysis

3.4.2 Uncertainty Analysis

The uncertainties of the resonance frequencies f_- and f_+ were obtained from Lorentzian fits to the inner dips close to the zero-field splitting and represent statistical uncertainties determined by the signal-to-noise ratio and the spectral resolution.

The derived parameters $D = (f_+ + f_-)/2$ and $E = (f_+ - f_-)/2$ inherit their statistical uncertainties from f_- and f_+ . In addition, systematic contributions arise from the determination of the applied magnetic field, which is generated by three orthogonal coil pairs and inferred from the applied currents. As discussed for the Zeeman splitting experiment, an effective uncertainty of the coil-generated

magnetic field was included and propagated to all field-dependent quantities.

Second-order polynomial fits were applied to $f_{\pm}(B)$ to describe the nonlinear Zeeman effect. The resulting uncertainties of the quadratic coefficients reflect both the limited magnetic-field range and the systematic uncertainty of the applied magnetic field.

3.5 Hyperfine Structure of the NV Centre

To resolve the hyperfine structure of the NV centre, ODMR spectra were recorded with the magnetic field aligned along a single NV axis. A microwave field was applied via the RF antenna while the microwave frequency was swept around the resonance.

The measurement was repeated for different microwave power settings (specified in dBm) in order to study the influence of the driving strength on the observed hyperfine features. For each power level, an ODMR spectrum was acquired under otherwise identical conditions.

The recorded spectra were analysed by identifying the hyperfine-resolved resonance dips and extracting their characteristic parameters for comparison between the different power settings.

3.5.1 Observed Hyperfine Structure

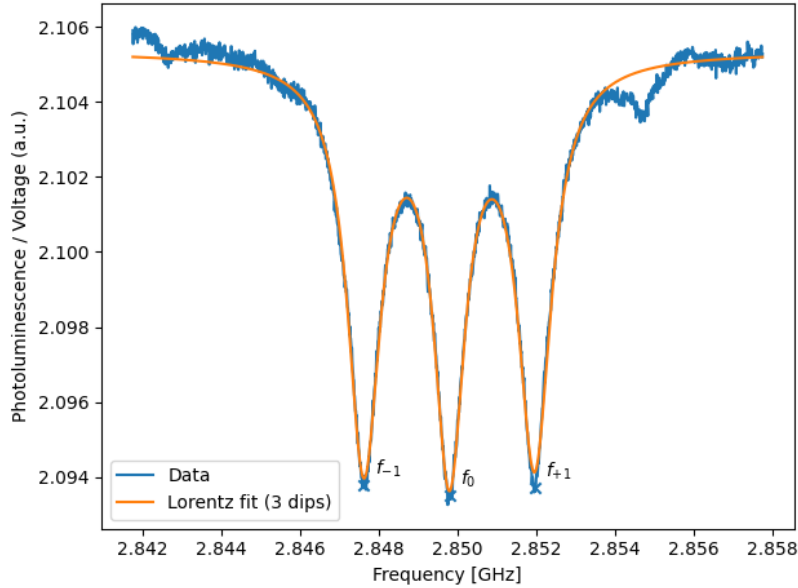


Figure 9: ODMR spectrum around the hyperfine triplet

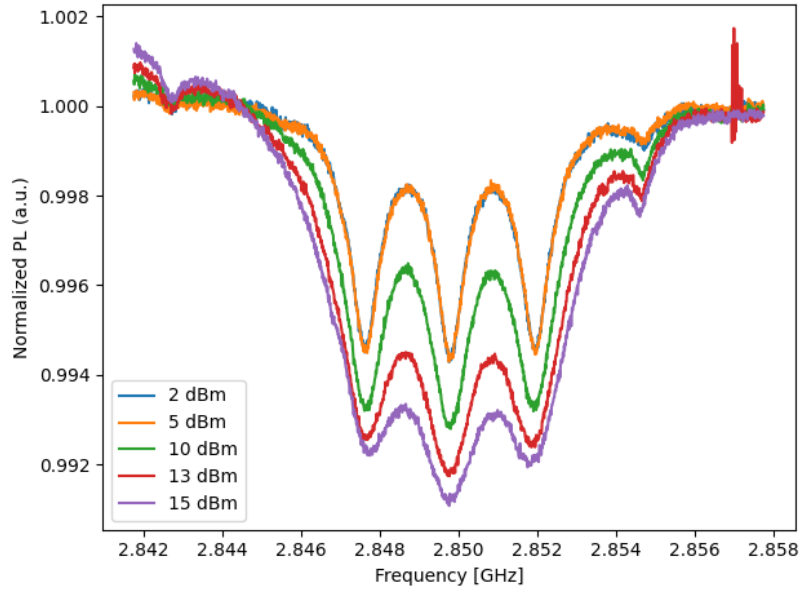


Figure 10: Normalized ODMR spectra around the hyperfine triplet for different RF powers. Increasing RF power leads to power broadening and changes in line contrast

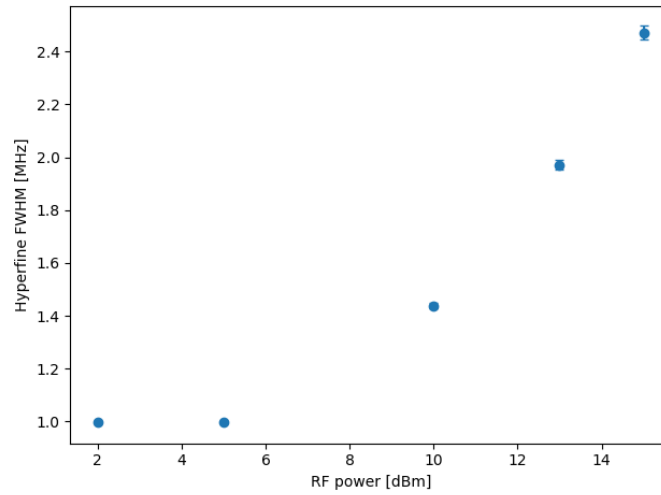


Figure 11: Mean hyperfine linewidth (FWHM) as a function of RF power

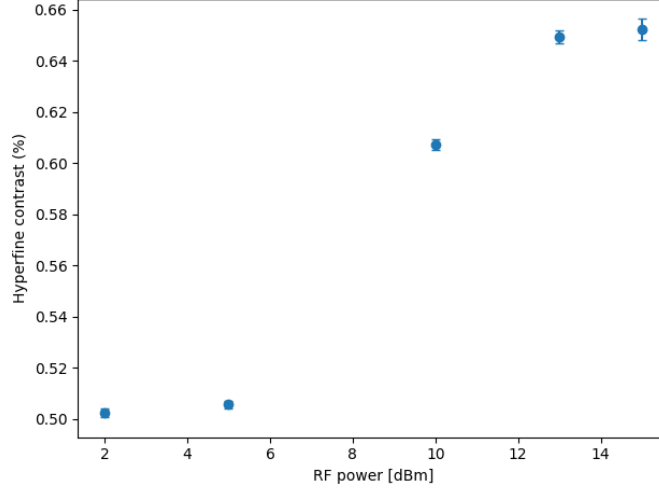


Figure 12: Mean hyperfine contrast as a function of RF power, obtained from the fitted dip amplitudes relative to the baseline signal

RF power [dBm]	f_{-1} [GHz]	f_0 [GHz]	f_{+1} [GHz]
2	2.847621 ± 0.003	2.849788 ± 0.003	2.851952 ± 0.003
5	2.847627 ± 0.002	2.849799 ± 0.002	2.851962 ± 0.002
10	2.847626 ± 0.004	2.849765 ± 0.004	2.851951 ± 0.004
13	2.847623 ± 0.007	2.849714 ± 0.007	2.851908 ± 0.009
15	2.847653 ± 0.010	2.849681 ± 0.010	2.851831 ± 0.016

Table 3: Extracted hyperfine resonance frequencies for different RF powers

RF power [dBm]	FWHM [MHz]	Contrast [%]	Δf_{HF} [MHz]
2	0.998 ± 0.006	0.502 ± 0.002	2.1656 ± 0.0020
5	0.997 ± 0.005	0.506 ± 0.001	2.1675 ± 0.0017
10	1.438 ± 0.010	0.607 ± 0.002	2.1622 ± 0.0032
13	1.971 ± 0.017	0.649 ± 0.003	2.1421 ± 0.0054
15	2.471 ± 0.025	0.652 ± 0.004	2.0890 ± 0.0086

Table 4: Mean linewidths (FWHM), contrast, and hyperfine splitting as a function of RF power

The ODMR spectrum around the hyperfine triplet is shown in Fig. 9. The three hyperfine resonances are clearly resolved and are well described by a sum of three Lorentzian dip functions, yielding the resonance frequencies f_{-1} , f_0 and f_{+1} . The extracted dip positions are stable within their statistical uncertainties, indicating a reliable determination of the hyperfine structure.

From the fitted dip positions, a mean hyperfine splitting of

$$\Delta f_{\text{HF}} = 2.16 \pm 0.01 \text{ MHz}$$

is obtained. This value is consistent with previously reported values for the NV centre, such as the hyperfine splitting $\Delta f_{\text{HF}} = 2.163 \pm 0.008 \text{ MHz}$ reported in the experimental manual.

Figure 10 compares normalized ODMR spectra recorded at different RF powers. With increasing RF power, the hyperfine resonances broaden visibly, while their central frequencies remain unchanged within uncertainty. This behaviour is characteristic of power broadening and reflects the increasing microwave driving strength.

The quantitative evolution of the mean hyperfine linewidth is shown in Fig. 11. A clear increase of the FWHM with RF power is observed, confirming power broadening of the hyperfine transitions. In contrast, the hyperfine splitting Δf_{HF} remains approximately constant over the investigated power range, as summarized in Table 4, indicating that the underlying hyperfine interaction itself is not affected by the applied RF power.

Figure 12 shows the corresponding hyperfine contrast as a function of RF power. The contrast increases monotonically with RF power, consistent with an enhanced driving efficiency of the spin transitions. At higher powers, the simultaneous increase in contrast and linewidth illustrates the typical trade-off between signal strength and spectral resolution in ODMR measurements.

3.5.2 Uncertainty of the Hyperfine Analysis

The uncertainties of the hyperfine parameters originate primarily from the Lorentzian fits used to extract the dip positions and linewidths. These statistical uncertainties reflect the finite signal-to-noise ratio and the local frequency resolution of the ODMR spectra. Systematic contributions related to the determination of the applied RF power and the coil-generated magnetic field may additionally affect the absolute values, but do not influence the observed trends with RF power. Overall, the quoted uncertainties adequately describe the experimental precision of the hyperfine analysis.

3.6 Determination of the Earth's Magnetic Field

The Earth's magnetic field was determined using optically detected magnetic resonance (ODMR) of an NV centre in diamond. A static magnetic field was applied along a selected coil axis by driving a current through the corresponding pair of coils, while the remaining coils were kept at zero current.

For each coil axis, two ODMR spectra were recorded: one with a positive coil current and one with a negative coil current of equal magnitude. Reversing the current direction inverts the contribution of the coil-generated magnetic field, while the Earth's magnetic field remains unchanged.

From the two spectra, the resonance frequencies were extracted and the corresponding Zeeman splittings were determined. By comparing the resonance positions obtained for positive and negative currents, the contribution of the Earth's magnetic field along the respective axis was isolated from the coil-generated field.

This procedure allows the determination of the Earth's magnetic field component along the selected axis without requiring an absolute calibration of the coil field strength.

3.6.1 Measured Earth's Magnetic Field

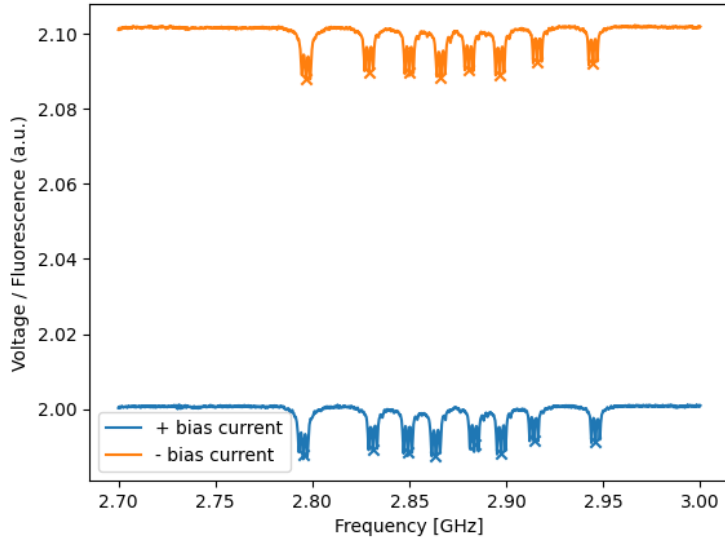


Figure 13: ODMR spectra recorded with positive and negative coil bias currents. Reversing the current polarity inverts the coil-generated magnetic field, while the contribution of the Earth's magnetic field remains unchanged

From two ODMR spectra recorded with equal-magnitude coil currents of opposite polarity, the projection of the Earth’s magnetic field along the selected coil axis was determined. Reversing the direction of the coil current inverts the contribution of the coil-generated magnetic field, while the Earth’s magnetic field remains unchanged. This enables a reliable separation of the geomagnetic field from the applied bias field without requiring an absolute calibration of the coil field strength.

The resulting Earth magnetic field vector was obtained as

$$\vec{B}_{\text{Earth}} = \begin{pmatrix} -32.7 \pm 0.1 \\ -31.1 \pm 0.1 \\ +9.3 \pm 0.1 \end{pmatrix} \mu\text{T},$$

corresponding to a total field magnitude of

$$|\vec{B}_{\text{Earth}}| = 46.1 \pm 0.1 \mu\text{T}.$$

The measured Earth magnetic field is consistent with the values reported in the experimental manual, where a field vector of

$$\vec{B}_{\text{Earth}} = \begin{pmatrix} -13.73 \pm 1.89 \\ 0.59 \pm 0.94 \\ -46.81 \pm 5.90 \end{pmatrix} \mu\text{T},$$

and a corresponding magnitude of

$$|\vec{B}_{\text{Earth}}| = 48.8 \pm 9.3 \mu\text{T}$$

are obtained. The agreement in the order of magnitude confirms the validity of the measurement approach.

3.6.2 Uncertainty of the Earth’s Magnetic Field Measurement

The uncertainties reported for the Earth’s magnetic field components are primarily determined by the precision with which the ODMR resonance positions can be extracted from the measured spectra. These uncertainties reflect the finite signal-to-noise ratio of the measurement and the limited spectral resolution, and are obtained directly from the Lorentzian fits used to determine the dip positions.

Beyond these fit-related contributions, additional uncertainties arise from the experimental realization of the magnetic field itself. The applied field is generated by three orthogonal pairs of coils and reconstructed from the corresponding coil currents using calibration factors. Imperfections in these calibration factors, geometric tolerances of the coil setup, and residual misalignment between the NV axis and the laboratory reference frame introduce systematic deviations that are not captured by the statistical fit uncertainties.

The experimentally obtained field magnitude

$$|\vec{B}_{\text{Earth}}| = 46.1 \pm 0.1 \mu\text{T}$$

is slightly smaller than the reference value reported in the experimental manual,

$$|\vec{B}_{\text{Earth}}| = 48.8 \pm 9.3 \mu\text{T}.$$

Since the observed difference exceeds the purely statistical uncertainty, it is attributed to systematic effects of the coil-generated field and its projection onto the NV axis. Nevertheless, the measured value lies within the expected range for the local geomagnetic field and confirms the overall consistency of the measurement.

3.7 Strong Magnetic Field

To demonstrate the response of the NV centre to strong external magnetic fields, a permanent magnet was manually brought close to the diamond sample and subsequently removed again. During this process, an ODMR spectrum was continuously recorded while the microwave frequency was swept.

3.7.1 Observed ODMR response in a strong magnetic field

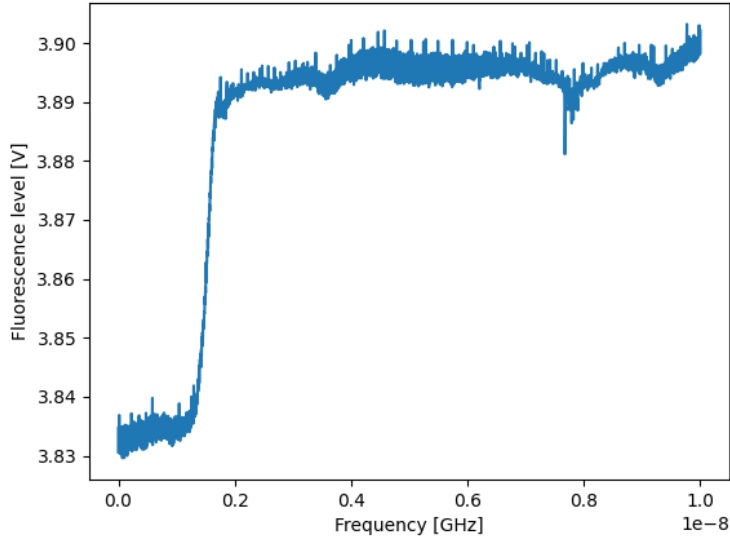


Figure 14: ODMR spectrum recorded in the presence of a strong external magnetic field applied by a permanent magnet

Figure 14 shows the ODMR spectrum recorded while a strong external magnetic field was applied by positioning a permanent magnet close to the diamond.

The ODMR dips are significantly shifted and broadened. When the magnet is removed, the spectrum returns to its original low-field structure. This behavior qualitatively demonstrates the strong Zeeman splitting induced by large magnetic fields and the sensitivity of the NV centre to changes in the local magnetic environment.

4 Appendix

List of Figures

1	Optical bench configuration used for the experiments on NV centers	3
2	ODMR spectrum recorded at 35% laser power.	7
3	ODMR spectrum recorded in the presence of an external magnetic field not aligned with a single NV axis	8
4	ODMR spectrum recorded with the magnetic field aligned along one NV axis	9
5	ODMR spectra recorded for different magnetic field strengths B (all measurement files)	11
6	Resonance frequencies f_+ and f_- as a function of the applied magnetic field B . Solid lines indicate linear fits used to extract the gyromagnetic ratio	12
7	ODMR spectra recorded at different magnetic field strengths in a narrow frequency range around the zero-field splitting	14
8	Resonance frequencies f_+ and f_- as a function of the applied magnetic field B	15
9	ODMR spectrum around the hyperfine triplet	16
10	Normalized ODMR spectra around the hyperfine triplet for different RF powers. Increasing RF power leads to power broadening and changes in line contrast	17
11	Mean hyperfine linewidth (FWHM) as a function of RF power	17
12	Mean hyperfine contrast as a function of RF power, obtained from the fitted dip amplitudes relative to the baseline signal	18
13	ODMR spectra recorded with positive and negative coil bias currents. Reversing the current polarity inverts the coil-generated magnetic field, while the contribution of the Earth's magnetic field remains unchanged	20
14	ODMR spectrum recorded in the presence of a strong external magnetic field applied by a permanent magnet	22

List of Tables

1	Extracted resonance frequencies and frequency splittings for different applied magnetic field strengths	12
2	Extracted resonance frequencies near the zero-field splitting and derived parameters for the nonlinear Zeeman analysis	15
3	Extracted hyperfine resonance frequencies for different RF powers	18
4	Mean linewidths (FWHM), contrast, and hyperfine splitting as a function of RF power	18